

A100-L400M30 Manual

RS485, 10km transmission , 410-490MHz LoRa spread spectrum

Star networking, auto-relay, AES encryption, ModBus protocol



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RS485, 10km, 410-490MHz LoRa spread spectrum, Star networking, auto-relay, AES encryption, ModBus protocol

1. Product overview

A100-L400M30 is a 410-490MHz, 1W, high stability, industrial grade wireless data transceiver. The module adopts LoRa spread spectrum technology and has RS485 interface. It adopts highly efficient cyclic error correction algorithm with high coding efficiency and strong error correction capability, which greatly improves the module's anti-interference and high stability.

2. Product features

- With data broadcast, data monitoring, target transmission, master-slave mode, automatic relay, target wake-up, automatic answer
- Auto-answer, the receiver automatically replies ACK to ensure transmission stability
- Receiving sensitivity up to -129dBm, transmission distance up to 10,000 meters^[1]
- Automatic relay, continuous transmission
- AES encryption
- Readable module RSSI^[2]
- Built-in multiple exception handling mechanisms to ensure long and stable operation of the module
- Maximum transmit power is 30dBm (1W)
- Multi-level baud rate^[1]
 - 8 common baud rates, default baud rate is 9,600bps
 - Baud rate range: 1,200bps-11,5200bps
- Multi-level air speed^[1]
 - 8 levels of adjustable air rate, default air rate 4.8kbps
- 1.2-62.5kbps
- Frequency 410-490MHz, providing 81 channels with 1MHz interval for each channel^[4]
 - 410MHz + CHAN * 1MHz
 - CHAN: 0x00-0x50H (corresponding to 410-490MHz)
- Operates in the 433MHz application-free band by default in China
- Supply voltage range
 - 8.0V-28VDC
- Built-in power management module to ensure stable power supply, able to meet a variety of system requirements
- 512 byte ring FIFO
- 512 bytes of send FIFO
- Internal automatic
- Internal automatic packet splitting for transmission
- Some combinations of airspeed and baud rate can send infinitely long packets

RS485 interface, 10km transmission , 410-490MHz LoRa spread spectrum Star networking, auto-relay, AES encryption, ModBus protocol

- Broadcast data and monitoring^[5]
 - The module address is set to 0xFFFF, which allows monitoring to data transmissions from all module on the same channel. The transmitted data can be received by any module with any address on the same channel
- Target transmission^[5]

- Data can be sent from any module and received from a specified module. When multiple receiving modules have the same address channel, data can be received.
- Data can be transmitted point-to-point across channels
- Can achieve a variety of applications such as networking and relaying
- Wake-on radio^[5]
- When the wireless wake-up time is not 0, the sender module will automatically add a wake-up code when sending data to wake up the target transmission module^[6]
- Same transmission method as target transmission

Remark.

[1] Test conditions: clear, open, maximum power, antenna gain 5dBi, height greater than 2m, 2.4k air rate

[2] Module RSSI is internal intelligent processing, the user does not need to care, special needs can be read by command

[3] See the SPEED register in Chapter 7, Section 7.2 Module Parameter Registers for details

[4] See the CHAN register in Chapter 7, Section 7.2 Module Parameter Register for details

[5] See Chapter 8 for details on module functions

[6] The DTU module is compatible with our UART module, and can wake up the UART module over the air.

3. Products series

Module Model	Working frequency (Hz)	Interface	Size (mm)	Maximum transmit power (dBm)	Communication distance (km)	Antenna Form
A100-L400M21	410M-525M	RS485	90*62*28	21	5	Glue stick, Magnetic antenna
A100-L433M30	410M-441M	RS485	90*62*28	30	10	Glue stick, Magnetic antenna

4. Electrical parameters

Conditions: Tc = 25°C, VCC = 12VDC

Module Model	Parameter Name	Description	Minimum value	Typical values	Maximum value	Unit
A100-L400M30	Supply voltage		8.0		28	VDC
A100-L400M30	Emission current			260		mA
A100-L400M30	Receiving current			6		mA
A100-L400M30	Working Frequency	410-525MHz, 116 channels, 1MHz step, factory default 433MHz	410	433	525	MHz
A100-L400M30	Transmitting power			30		dBm
A100-L400M30	airspeed	8 levels adjustable (1.2kbps, 2.4kbps, 4.8kbps, 9.6kbps, 19.2kbps, 38.4kbps, 50kbps, 62.5kbps)	1.2k	4.8k	62.5k	bps
A100-L400M30	Reception sensitivity	Receiving sensitivity is independent of serial port baud rate and delay time		-129		dBm
A100-L400M30	Operating temperature	Industrial Products	-40		+85	°C
A100-L400M30	Working humidity	Relative humidity, non-condensing	10%		90%	
A100-L400M30	Storage temperature		-40		+125	°C

5. Module connection method

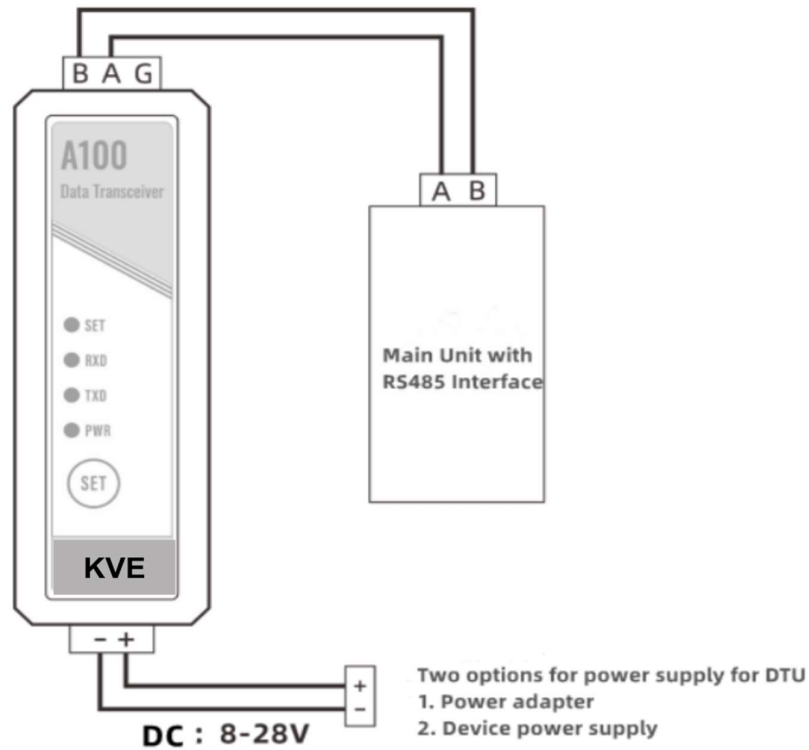


Figure 5-1 Recommended connection diagram

Note: [1] See DTU product manual for detailed connection methods and procedures

6. Module reset

The default state of the module RESET indicator is off. Press and hold the RESET button until the RESET indicator flashes, release the button, then the device will produce a reset and restore the factory settings.

7. Module commands

7.1 Command format

The parameters of our DTU can be configured to support read and write command operation, and the detailed commands are as follows.

➤ Configuration mode command overview table

The parameter configuration command only supports parameter modification in the configuration mode, and is valid when the RESET lamp is lit, at which time the serial port parameters are 9600, 8N1.

The configuration command is implemented by adding the register address and register length to the command word, which can realize the configuration of multiple configuration items continuously at the same time.

The command format is as follows.

Command word	Register start address	Number of configuration registers	Register Value	Register Value	Register value n
CMD	REG	LEN	DATA0	...	DATAn
CMD [7] 0 for read, 1 for write	The range of REG in this model is 0x00~0x21	Indicates the number of registers to be operated. When the number of registers to be operated exceeds the number of current operation rights, an error will be returned, for example, if a read-only register is written, an error will be returned.	When reading, you can leave it unfilled, when writing, make sure to correspond to the register length and total length, otherwise it will report an error		
CMD [6] 0 for local command, 1 for remote configuration					
CMD [5]. 0 is return operation success, 1 is return operation error					
CMD[4...0] Reserved					

Note: Register addresses are distinguished by function, not by occupied bytes

Command functions are detailed in the following table, using the default factory configuration as an example.

Command Format	Module Answer	Description
0x80 ADDR LEN VAL1 VAL2 ... VALn (See parameter configuration register description for details)	0x80 ADDR LEN	Configuration success, configuration parameters power down and save
	0xC0 ADDR LEN	Configuration failed, the original configuration parameters were not changed
0x80 0x0B 0x01 0x77 0x77 0x77 0x2E 0x61 0x73 0x68 0x69 0x6E 0x69 0x6E 0x67 0x2E 0x63 0x6F 0x6D	0x80 0x0B 0x01	Data key success
	0xC0 0x0B 0x01	Data key failure
0x00 0x04 0x1C	0x00 0x00 0x25 0x80 0x00 0x02 0xBD 0x00 0x01 0x05 0x03 0xE8 0x00 0x77 0x77 0x77 0x77 0x2E 0x61 0x73 0x68 0x69 0x6E 0x69 0x6E 0x67 0x2E 0x63 0x6F 0x6D 0x7C 0x7C 0x7C 0x7C 0x05 0x40 0x00 0x23 0x00 0x00 0x00 0x00 0x3C 0x3C 0x00 0x0A 0x04 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00	The module returns the parameters of all currently writable registers in hexadecimal format
0x00 0x00 0x01	A100-L400M30-v1.2.00	The module will return the current version number in ASCII format
0x80 0x22 0x01	OK	The module generates a reset, during which the module performs a self-test and the AUX goes low, after the reset the AUX goes high and the module starts to work normally. At this time, you can switch the status or initiate

		the next command
0x80 0x23 0x01	OK	Restore default parameters configuration successfully
0x00 0x01 0x01	XXXX	Read the current data signal RSSI and environment signal RSSI directly. e.g. the module returns XXXX, the first two XX represent the current environment RSSI, the last two XX represent the current data RSSI, convert them to decimal data, it means the current signal strength. This command can be used in all modes

➤ Operating mode command overview table

To configure parameters in operating mode, the configuration command needs to satisfy the following format.

Sync header + ID code + header + command code inverse + command code + data (length is not fixed, refer to command overview table) + CS + sync header inverse + ID code inverse + header inverse + end code

Number of data bits	Value	Description
0, 1	0xAF 0xAF	2BYTE Sync Head
2, 3	0xFF 0xFF	2BYTE ID code, can be a random number, default 0x55 0x55
4	0x5A	1BYTE header
5	0xFF	1BYTE Command Code Inversion (Command Code Reference Command Overview Table)
6	0xFF	1BYTE command code (command code reference command overview table)
7	0xFF	1BYTE start address
8	0xFF	1BYTE command length
8~n	0xFF	Command data, refer to the parameter register table
n+1, n+2	0xFF 0xFF	2BYTE check code, the high eight bits are the sum of all data before CS to take the low 8 bits, the low eight bits are all data before CS iso-or
n+3, n+4	0x50 0x50	2BYTE synchronous header reversal
n+5, n+6	0xFF 0xFF	2BYTE ID code is inverted, default 0xAA 0xAA
n+7	0xA5	1BYTE header is reversed
n+8, n+9	0x0D 0x0A	2BYTE end code

The working mode command function is detailed in the following table, taking the default factory configuration as an example.

Command Format	Module Answer	Description
AF AF 55 55 5A FF 00 04 1C 81 BD 50 50 AA AA A5 0D 0A	00 00 25 80 00 02 FA 00 01 05 03 E8 00 77 77 77 77 2E 61 73 68 69 6E 69 6E 67 2E 63 6F 6D 7C 7C 7C 7C 7C 7C 05 40 00 23 00 00 00 00 3C 3C 00 0A 19 00 80 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	The module returns the parameters of all currently writable registers in hexadecimal format
AF AF 55 55 5A 7F 80 04 1C 00 00 25 80 00 02 FA 00 01 05 03 E8 00 77 77 77 77 77 2E 61 73 68 69 6E 69 6E 67 2E 63 6F 6D 7C 7C 7C 7C 7C 7C 05 40 00 23 00 00 00 3C 3C 00 0A 19 00 80 00 00 00 00 00 00 00 00 00 00 00 00 00 53 8D 50 50 AA AA A5 0D 0A	80 04 1C	Write all parameters

7.2 Module parameter register

Parameter registers (parameter registers cannot be used alone, they must be used in accordance with the command format of the configuration parameters)

15	0x0E	R/W	1	Serial packet length in bytes
16	0x0F	R/W	2	Serial port packing cycle in 0.1ms
17	0x10	R/W	2	Reserved
18	0x11	R/W	1	Reserved
19	0x12	R/W	1	Reserved
20	0x13	R/W	1	Reserved
21	0x14	R/W	1	Wake-up time in 100ms units
22	0x15	R/W	1	Reserved
23	0x16	R/W	1	Data window length in polling mode and active reporting mode, in 200ms
24	0x17	R/W	2	Job Options Bit9 turns on data sending before scanning for new to idle Bit8 opens data sending with random delay before sending Bit6 opens the group number output to request lost packets after the reception is complete Bit5 opens the function of group packet output Bit4 turns remote configuration on or off Bit3 When configured over the air, whether to reply to ACK, only available in one-to-one configuration Bit2 When configured over the air, whether to output the result from the serial port Whether to output packet and packet separator on Bit1 serial output Bit0 output address
25	0x18	R/W	1	Local Group Number
26	0x19	R/W	1	Local Address
27	0x1A	R/W	1	Fixed group number
28	0x1B	R/W	1	Destination Address
29	0x1C	R/W	1	In relay mode, the path A group number
30	0x1D	R/W	1	Relay mode, path A address
31	0x1E	R/W	1	In relay mode, the path B group number
32	0x1F	R/W	1	Relay mode, path B address
33	0x20	X	1	Module reboot
34	0x21	X	1	Module restores factory settings

7.3 Module factory settings

Register factory configuration table

Serial number	Addresses	Register Permissions	Register Length	Default parameters
1	0x00	R	20	A100-L400M30-V1.2.00, module version number
2	0x01	R	4	0x0000 (default), which means that the current environment RSSI is 0x00 and the current package RSSI is 0x00
3	0x02	R	2	Reserved
4	0x03	R	2	Reserved
5	0x04	R/W	4	0x00002580 (9600bps, default) serial port baud rate of the module, customizable, range 1200~115200
6	0x05	R/W	1	0x00 (default) Bit4 is 0. Indicates that the data bits are 8 bits Bit5 is 0. Indicates that the stop bit is 1 bit

				Bit2 Bit1 is 00 means no checksum Serial port parameters are 8-bit data bit 1-bit stop bit No parity bit
7	0x06	R/W	2	0x02FC (default) Bit10 Bit9 Bit8 Bit7 Bit6 Bit5 is the channel number, the default is 10111 (23) channels Bit4 Bit3 is the power number, the default is 11, the power is 30dBm Bit2 Bit1 Bit0 is the null speed number, default is 010, the null speed is 2.4K
8	0x07	R/W	2	0x0001 (default) Operating mode is transparent transmission
9	0x08	R/W	1	0x05 (default) Bit8 is 0 to turn off stable transmission by default Bit7~Bit0 is 5, which means the default number of r-transmission is 5 times
10	0x09	R/W	2	1000 in ms, stable transmission re-transmission interval is 1S by default
11	0x0A	R/W	1	0x00 (default) Bit0=0 means that when in master-slave mode, as the host Bit1=0 means that when in polling mode, as a slave Bit2=0 means that when it is in the timed-up report, it acts as a slave
12	0x0B	R/W	16	0x77 0x77 0x77 0x2E 0x61 0x73 0x68 0x69 0x6E 0x69 0x6E 0x67 0x2E 0x63 0x6F 0x6D (default) Indicates that the packet is encrypted using the above AES key
13	0x0C	R/W	5	(default) When the separator output is turned on, the above separator is output between packages
14	0x0D	R/W	1	0x05 (default) Indicates that the maximum valid length for using separators is 5
15	0x0E	R/W	1	64 (default) When the serial input is more than 64 bytes, then every 64 bytes will be treated as 1 packet
16	0x0F	R/W	2	35 (default) When the serial port data idle time exceeds 3.5ms, then the currently received data is treated as a packet
17	0x10	R/W	2	Reserved
18	0x11	R/W	1	Reserved
19	0x12	R/W	1	Reserved
20	0x13	R/W	1	Reserved
21	0x14	R/W	1	0 (default) No wake-up code and no sleep will be sent by default
22	0x15	R/W	1	Reserved
23	0x16	R/W	1	25 (default) Each window time is 5S
24	0x17	R/W	1	0x04 (default) Turn off wireless configuration function, turn off wireless reply to remote configuration result, turn on serial reply to wireless configuration result, turn off outputting separator between packets. Turn off outputting addresses.
25	0x18	R/W	1	0x00, the local group number is 0x00
26	0x19	R/W	1	0x00 with a local address of 0x00
27	0x1A	R/W	1	0x00, the Fixed group number is 0x00
28	0x1B	R/W	1	0x00, the destination address is 0x00
29	0x1C	R/W	1	0x00, in relay mode, the path A group number is 0x00
30	0x1D	R/W	1	0x00, in relay mode, the path A address is 0x00
31	0x1E	R/W	1	0x00, in relay mode, the path B group number is 0x00
32	0x1F	R/W	1	0x00, in relay mode, the path B address is 0x00

Remark.

- 1、 The configuration of the module parameters can use the parameter configuration software ASDS provided by the company, the configuration process and steps are detailed in the DTU product manual
- 2、 Fast communication test of the module and detailed procedures are described in the DTU product manual

8. Module function

8.1 Module features overview

Module menu			
Module Function	Data format of the sender	Data format of the receiver	Function introduction
Transparent broadcasting	User data	User data	Any station that sends data can be received by stations with the same address and the same channel. Sending data is transparent, and what you send is what you receive.
Target transmission	Receiver address + receiver channel + user data	User data	Data can be sent from any module station and received by specified module stations, and data can be received when multiple receiving module stations have the same address channel. Data can be transmitted point-to-point across channels.
Broadcast and monitoring (under target transmission)	0xFF+0xFF+receiver channel+user data	User data	Broadcast: any station sends data, and all stations under the same channel can receive data at the same time. Monitoring: a module with a 0xFFFF address can receive data sent by a module with any address under the same channel.
Wake-on radio	1. In transparent transmission mode. (automatically adds wakeup code 010101010 -----) + User Data 2. In target transmission mode. (automatically adds wakeup code 010101010 -----) + Receiver address + receiver channel + user data	User Data	1、 The station will automatically add a wake-up code when sending data 2、 The length of the wake-up code can be set by setting the wireless wake-up time

8.2 Module function detailed explanation

8.2.1 Transparent Broadcasting

Function description

Any station sends data, and any station with the same address and the same channel can receive data at the same time. Data is sent and received transparently, and what is sent is what is received.

Module settings

1. Bit 7 of the OPTION special function register needs to be configured to 0. Transparent transmission mode
2. Sending and receiving module addresses are set to the same value
3. The channels of the transmitting and receiving module are set to the same value

• Examples:

Transmitter		Receiver	
Module Address	0x1234 (factory default)	Module Address	0x1234 (factory default)
Module Channels	0x17 (factory default)	Module Channels	0x17 (factory default)
Sending data	User Data	Output Data	User Data
	0x11 0x22 0x33		0x11 0x22 0x33

8.2.2 Reliable Transmission

Function description

After the module sends the data, it will wait for the answer packet from the receiver. When the packet is received, it means the current packet is sent successfully, if the timeout is not received, then the sender module will automatically resend this packet data until the answer packet is received or until the maximum number of packets is sent.

Module settings

1. Bit 7 of the 0x08 special function register needs to be configured to 1 to turn on the stable transfer function, which is off by default.
2. It is necessary to configure bits 6~0 of the 0x08 special function register to the number of times re-transmission is required, maximum 127 times, default 5 times
3. The 0x09 special function register needs to be configured to require a re-transmission interval in ms, the default is 1000ms

• Examples:

Transmitter		Receiver	
Module Address	0x1234 (factory default)	Module Address	0x1234 (factory default)
Module Channels	0x17 (factory default)	Module Channels	0x17 (factory default)
Sending data	User Data	Output Data	User Data
	First send 0x11 0x22 0x33		
	No ACK received		
	nth send 0x11 0x22 0x33		0x11 0x22 0x33
	Receive ACK		

8.2.3 Master-slave mode

Function description

The master can send data to the specified slaves, and all slaves can receive the data sent by the master.

Module settings

1. MD0 = 0, MD1 = 0
2. Need to configure 0x07 special function register to 0x0004, master-slave mode
3. The addresses of the transmitter and receiver modules can be set differently
4. The channels of the transmitter and receiver modules are set to the same

• Examples:

Transmitter		Receiver	
Target group number address	0xXXXX	Local group number address	0x5678
Modular channels	0x18	Modular channels	0x18
Master-Slave	Mainframe	Master-Slave	From the machine
Sending data	Receive address high + receive address low + data	Output Data	User Data
	0x56 0x78 0x11 0x22 0x33		0x11 0x22 0x33

8.2.4 Target transmission

Function description

Data can be sent from any station and received from a specified station. When multiple receiving stations have the same address channel, data can be received.

Module settings

5. Bit 7 of the OPTION special function register needs to be configured to 1, target transmission mode
6. The addresses of the transmitting and receiving stations can be set differently
7. The channels of the transmitting and receiving module can be set differently
8. The first 3 bytes of the sender's data must be the receiver's address high + address low + channel

• Examples:

Transmitter		Receiver	
Module Address	0xFFFF	Module Address	0x5678
Module Channels	0xFFFF	Module Channels	0x18
Sending data	Receive address high + receive address low + receive channel + data	Output Data	User Data
	0x56 0x78 0x18 0x11 0x22 0x33		0x11 0x22 0x33

8.2.5 Broadcast under target transmission

Function description

Any station sends data, and all stations under the same channel can receive data.

Module settings

1. Bit 7 of the OPTION special function register needs to be configured to 1, target transmission mode
2. The addresses of the transmitting and receiving stations can be set differently
3. The channels of the transmitting and receiving module can be set differently
4. The first 3 bytes of the sender's data must be 0xFF+0xFF+receiver channel

• Examples

Transmitter		Receiver	
Module Address	0xFFFF	Module Address	0xFFFF
Module Channels	0xFFFF	Module Channels	0x17 (factory default)
Sending data	0xFF+0xFF+receive channel+user data	Output Data	User Data
	0xFF 0xFF 0x17 0x11 0x22 0x33		0x11 0x22 0x33

8.2.6 Monitoring under target transmission

Function description

Any module with a 0xFFFF address can receive data from a module with any address on the same channel.

Module settings

1. Bit 7 of the OPTION special function register needs to be configured to 1, target transmission mode
2. The address of the monitoring module must be set to 0xFFFF
3. The channels of the monitoring and transmitting stations must be set to the same
4. The first 3 bytes of the sender's data must be 0xFF+0xFF+monitoring channel

Examples:

Transmitter		Receiver	
Module Address	0x1234 (factory default)	Module Address	0xFFFF
Module Channels	0x17 (factory default)	Module Channels	0x17 (factory default)
Sending data	arbitrary address high + arbitrary address low + receive channel + user data	Output Data	User Data
	0xFF 0xFF 0x17 0x11 0x22 0x33		0x11 0x22 0x33

8.2.7 Relay

Function description

The device in relay mode will forward the received data backward according to the set path until the Fixed device for the purpose of extending the transmission distance.

Module settings

1. MD0 = 0, MD1 = 0. MD0 = 1, MD1 = 0
2. Need to configure 0x07 special function register to 0x0020, relay mode
3. Set the previous group number address and the next group number address in relay mode (0x1B~0x1E registers)
4. The channels of the relay module and the transmitter module must be set to the same

8.2.8 Output address

Function description

When the output address function is turned on, it precedes the user data with the address information of the sender so that the receiver can distinguish the source of the data.

Module settings

Need to turn on the 0x17 special function register to output address function

Examples:

Transmitter		Receiver	
Destination group number address	0x0000 (factory default)	Local group number address	0x0001 (factory default)
Modular channels	0x17 (factory default)	Modular channels	0x17 (factory default)
Sending data	User Data	Output Data	User Data
	0x11 0x22 0x33		0x00 0x00 0x11 0x22 0x33

8.2.9 Output separator

Function description

When the output delimiter function is turned on, it precedes the user data with a user-set delimiter so that the recipient can distinguish the boundaries of the data.

Module settings

1. Need to turn on the 0x17 special function register for output separator
2. Set the 0x0C special function register to the content of the separator you want to send
3. Set the 0x0D special function register to the length of the desired separator

Examples:

Sender		Receiver	
Destination group number address	0x0000 (factory default)	Local group number address	0x0001 (factory default)
Modular channels	0x17 (factory default)	Modular channels	0x17 (factory default)
Sending data	User Data	Output Data	User Data
	123456789		123456789

8.2.10 Configurable packet size

Function description

www.khangvietelectric.com

The data received from the serial port will not be sent out immediately, it will wait until a group packet boundary set by the user before passing the data out. The maximum group packet can be matched with 230 bytes, and if the packet splitting time is less than the transmission time of three bytes at the current baud rate, the module will automatically correct to the transmission time of three bytes.

- Module settings

1. The 0x0E special function register needs to be configured with the desired number of bytes of sub-packet length, 64 bytes by default.
2. The 0x0F special function register needs to be configured for the desired packet splitting interval, which defaults to three bytes of transmission delay.

8.2.11 Passing mode

- Function description

A device in pass-through mode will automatically transmit data down the line until the final node outputs data. If you need to output data at every node, just set the destination address to 0xFF.

Module settings

1. The 0x07 special function register needs to be configured to pass the mode 0x0100.
2. Set the address of the module at each level, and the address increases in sequence.

8.2.12 Send after random delay

Function description

A device that sends with random delay turned on will delay the data for a random period of time before sending it out each time, in order to reduce the chance of data collisions, and the maximum delay time will be less than the transmission time of a full packet.

Module settings: Need to turn on the 0x17 special function register for random delay

8.2.13 Send after scanning channel

Function description

The device that sends after opening the scanning channel will scan whether the current channel is occupied before sending data each time, and if it is occupied will wait for a period of time and then scan again until the channel is clean before sending the data out, in order to reduce the chance of data collision.

Module settings: It is necessary to turn on the 0x17 special function register to scan the channel and then send

8.2.14 Packet output

Function description

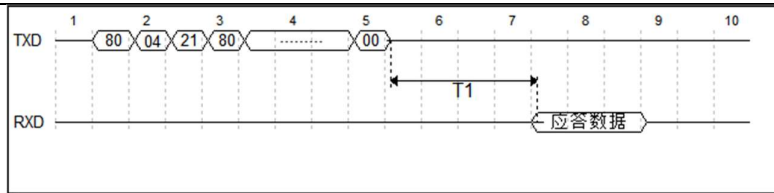
This function requires both the sender and the receiver to turn on the packet grouping function. When the receiver receives the data, it will group the data into a large packet, up to 800 bytes, and output a large packet all at once, which is more widely applicable.

- Module settings

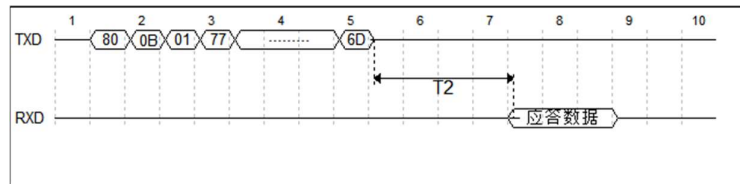
1. The 0x17 special function register needs to be turned on for packet output function for both transmit and receive
2. If you need to ensure the reliability of the data, you can choose to turn on the option to request lost packets after reception is complete under the 0x17 special function register.

9. Timing chart

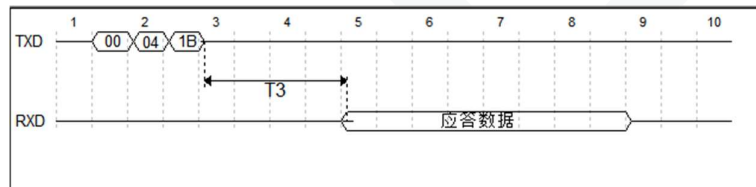
1. Module parameter configuration commands



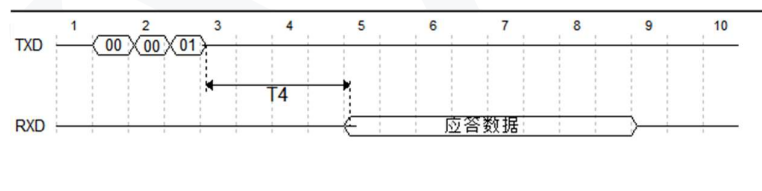
2. Module encryption key setting command



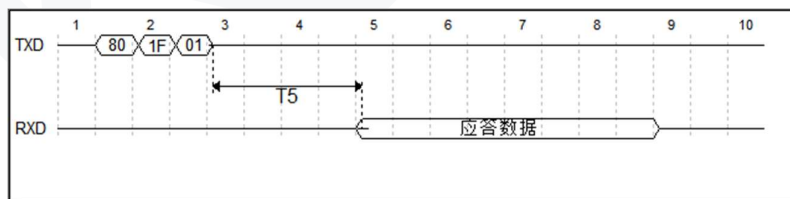
3. Read module configuration parameters command



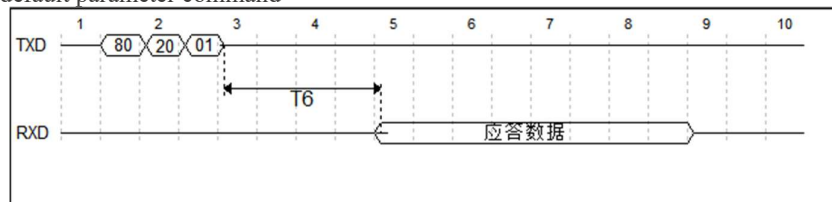
4. Read module hardware version number command



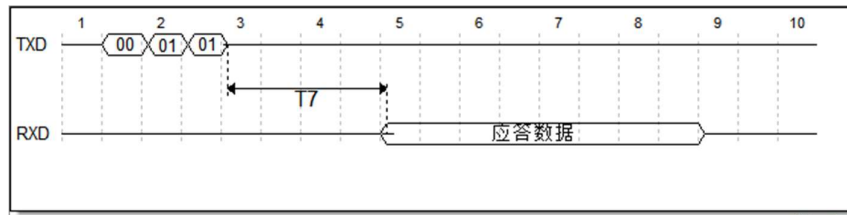
5. Module reset command



6. Restore module default parameter command



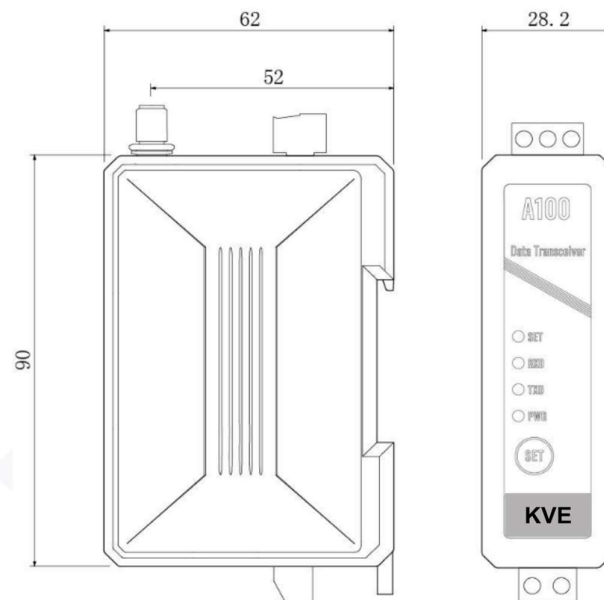
7. Read current data signal RSSI and environment RSSI commands directly



Parameter name	T_answer	Description	Minimum value	Typical values	Maximum value	Unit
Module command answer delay	T1	Parameter configuration delay		100		ms
	T2	Read module configuration parameters delay		40		ms
	T3	Read module hardware version number delay		100		ms
	T4	Waiting for module reset delay		40		ms
	T5	Read module voltage delay		60		ms
	T6	Configure module encryption key delay		60		ms
	T7	Recovery module default parameter delay		50		ms
Waiting for data transmission completion delay	T_Packet	Delay time required to finish sending a packet of data				ms

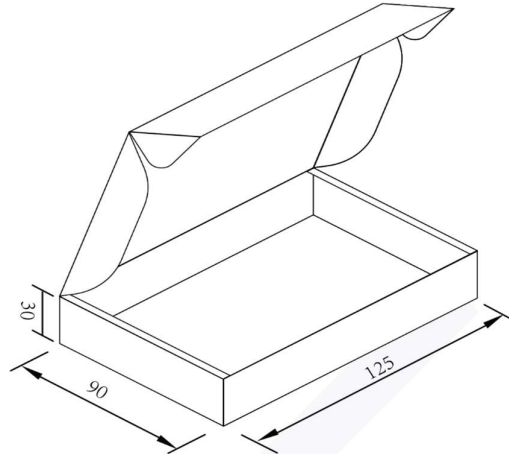
10. Package information

10.1 Mechanical size(mm)



11. Packaging method

11.1 DTU tooling(unit:mm)



Important notes and disclaimers

Due to the continuous improvement of hardware and software of the product, this specification is subject to change, and the latest version of the specification shall prevail in the end.

Users who use this product need to pay attention to the product dynamics at the official website so that users can get the latest information of this product in time.

The pictures and diagrams used in this specification are for illustrating the functions of this product and are for reference only. The measurement data in this specification are measured by our company at room temperature for reference only, please refer to the actual measurement.

Chengdu Ashining Technology Co., Ltd. reserves the right of final interpretation and modification of all contents in this specification.